

3-1 (a) Formulate node-voltage equations for the circuit in Figure P3-1. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.

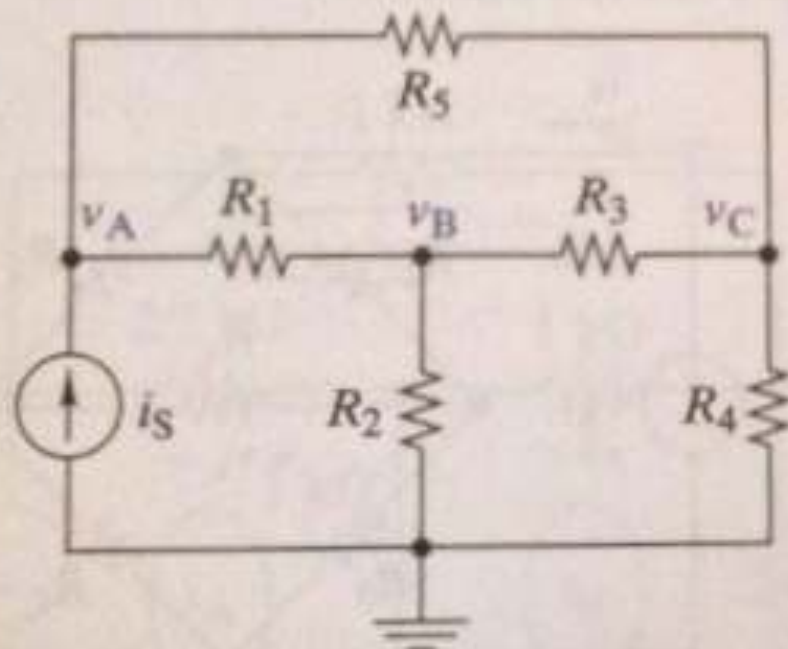


FIGURE P3-1

- 3-3 (a) Formulate node-voltage equations for the circuit in Figure P3-3. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve these equations for v_A and v_B .
(c) Use these results to find v_x and i_x .

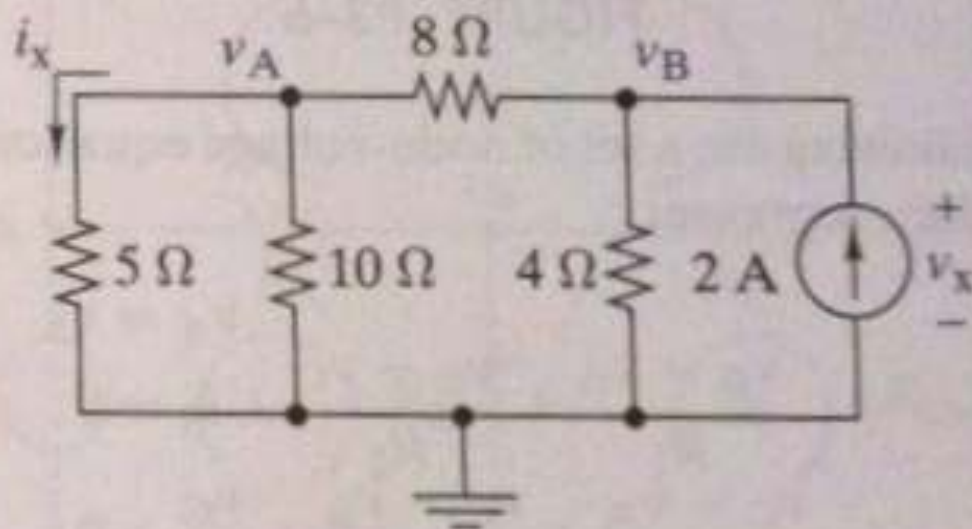


FIGURE P3-3

3-4 (a) Formulate node-voltage equations for the circuit in Figure P3-4.

(b) Solve these equations for v_A and v_B .

(c) Use these results to find v_X and i_X .

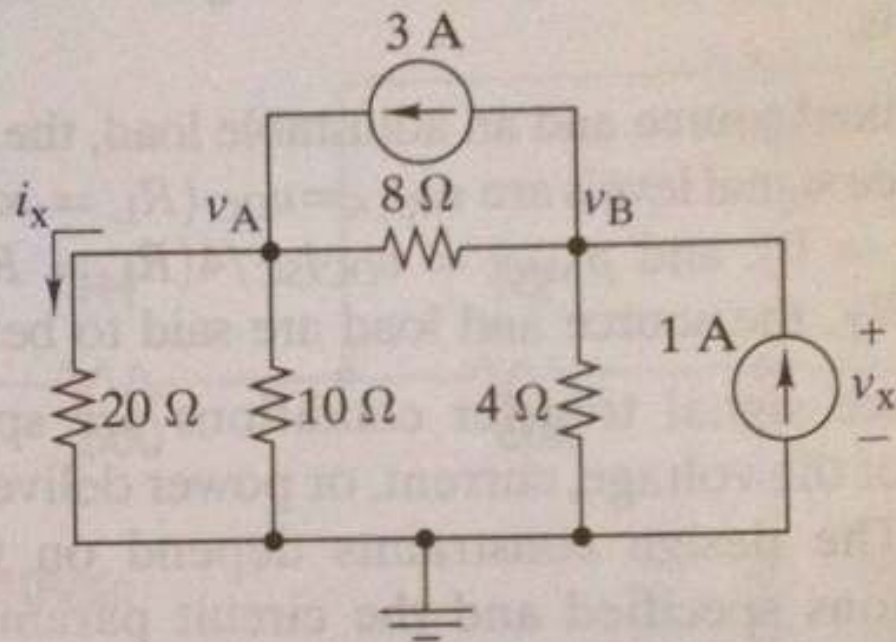


FIGURE P3-4

- 3-5 (a) Formulate node-voltage equations for the circuit in Figure P3-5. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
 (b) Solve these equations for v_A and v_C .
 (c) Use these results to find v_x and i_x .

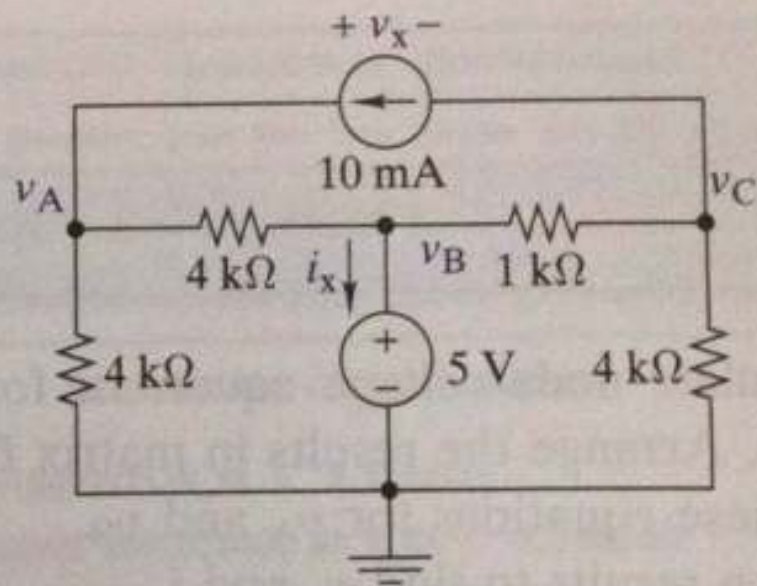


FIGURE P3-5

- 3-13(a) Formulate mesh-current equations for the circuit in Figure P3-13. Arrange the results in matrix form $\mathbf{Ax} = \mathbf{b}$.
(b) Solve for i_A and i_B .
(c) Use these results to find v_x and i_x .

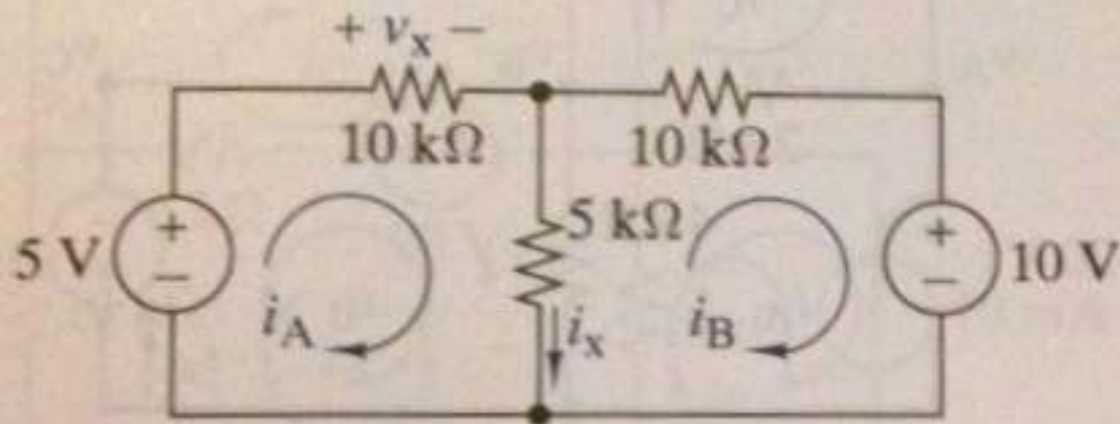


FIGURE P3-13

- 3-17 (a) Formulate mesh-current equations for the circuit in Figure P3-17.
- (b) Formulate node-voltage equations for the circuit in Figure P3-17.
- (c) Which set of equations would be easier to solve? Why?
- (d) Using MATLAB, find v_x and i_x in terms of the mesh-current variables.
- (e) Using MATLAB, find v_x and i_x in terms of the node-voltage variables.

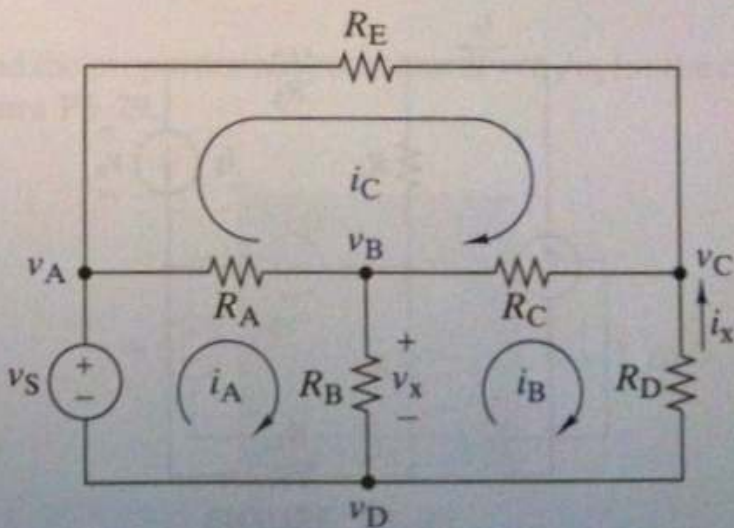


FIGURE P3-17

- 3-22(a)** Formulate mesh-current equations for the circuit in Figure P3-22.
- (b)** Formulate node-voltage equations for the circuit in Figure P3-22.
- (c)** Which set of equations would be easier to solve? Why?
- (d)** Find v_x and i_x using whichever method you prefer.

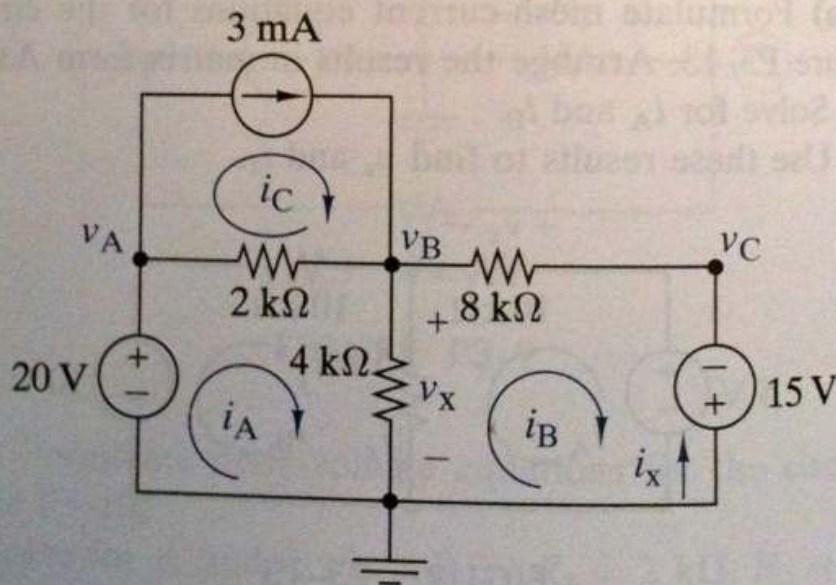


FIGURE P3-22

3-47 For the circuit in Figure P3-47 find its Thévenin and Norton equivalent circuits.

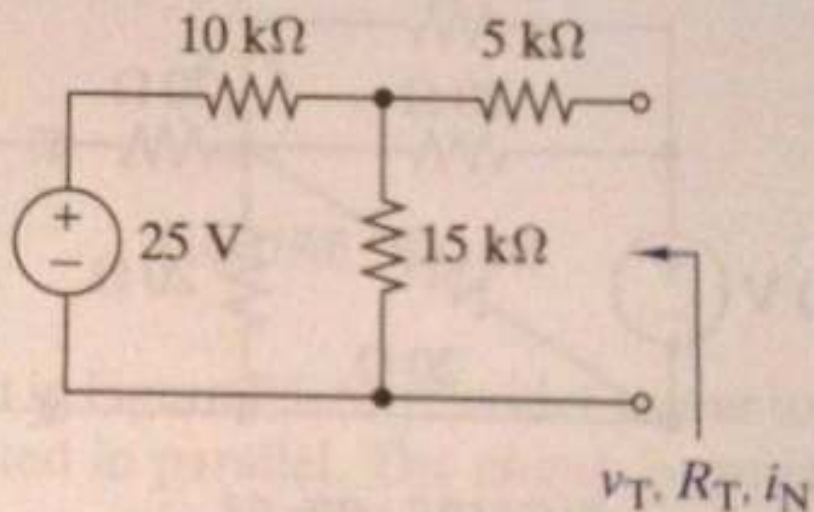


FIGURE P3-47

3-52 Find the Norton equivalent seen by R_L in Figure P3-52. Find the current through the load for when $R_L = 4.7\text{ k}\Omega$, $15\text{ k}\Omega$, and $68\text{ k}\Omega$.

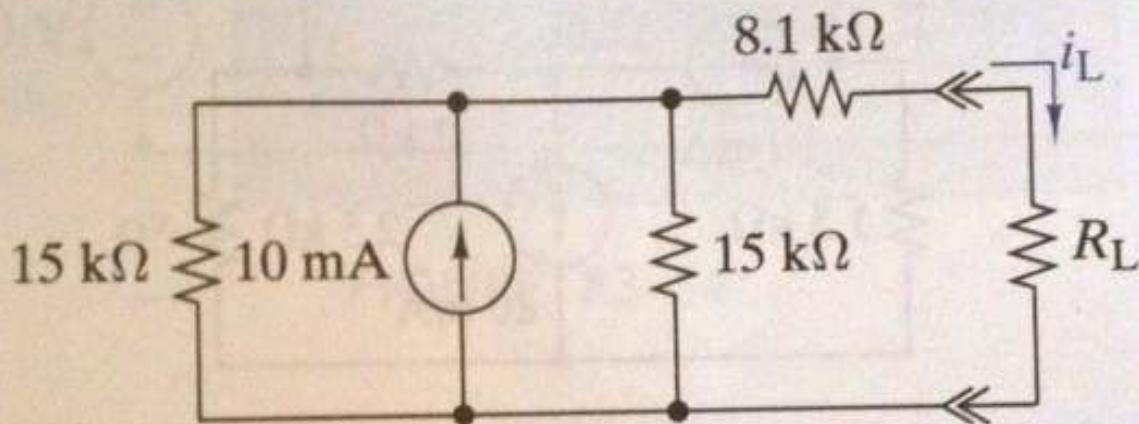


FIGURE P3-52

3-58 Assume that Figure P3-58 represents a model of the auxiliary output port of a car. The output current is $i = 1$ A when $v = 0$ V. The output voltage is $v = 12$ V when a $20\text{-}\Omega$ resistor is connected between the terminals. Suppose you wanted to charge a 12-V battery by connecting the battery at the port. How much current would the port deliver to the battery?

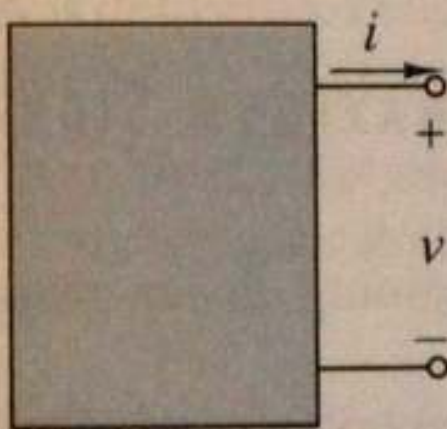


FIGURE P3-58